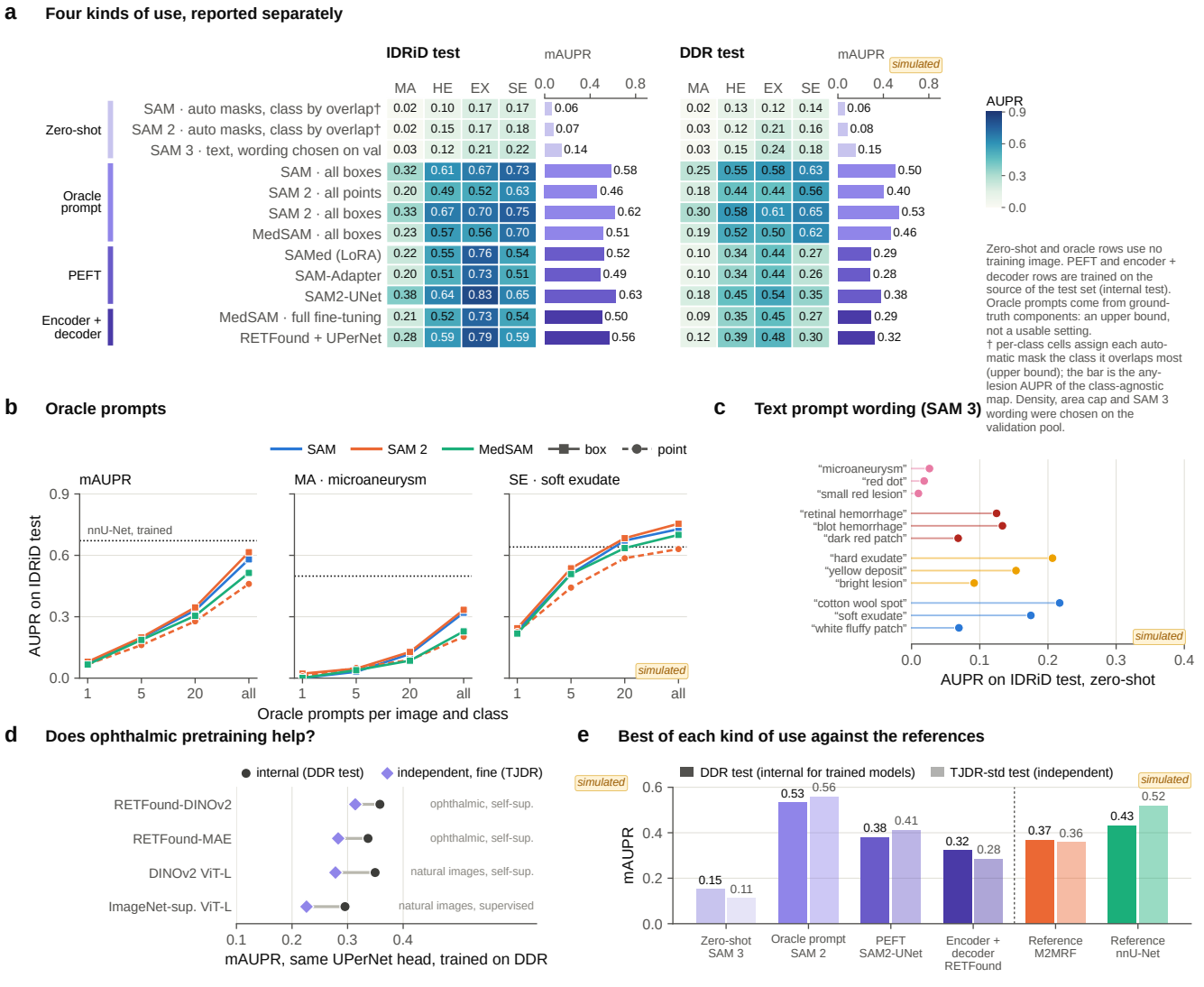
a
Four kinds of use, reported separately
IDRiD test
DDR test
mAUPR
simulated
MA HE EX SE
0.0 0.4 0.8
Zero-shot
Oracle prompt
PEFT
Encoder + decoder
SAM · auto masks, class by overlap† 0.02 0.10 0.17 0.17 0.06 | 0.02 0.13 0.12 0.14 0.06
SAM 2 · auto masks, class by overlap† 0.02 0.15 0.17 0.18 0.07 | 0.03 0.12 0.21 0.16 0.08
SAM 3 · text, wording chosen on val 0.03 0.12 0.21 0.22 0.14 | 0.03 0.15 0.24 0.18 0.15
SAM · all boxes 0.32 0.61 0.67 0.73 0.58 | 0.25 0.55 0.58 0.63 0.50
SAM 2 · all points 0.20 0.49 0.52 0.63 0.46 | 0.18 0.44 0.44 0.56 0.40
SAM 2 · all boxes 0.33 0.67 0.70 0.75 0.62 | 0.30 0.58 0.61 0.65 0.53
MedSAM · all boxes 0.23 0.57 0.56 0.70 0.51 | 0.19 0.52 0.50 0.62 0.46
SAMed (LoRA) 0.22 0.55 0.76 0.54 0.52 | 0.10 0.34 0.44 0.27 0.29
SAM-Adapter 0.20 0.51 0.73 0.51 0.49 | 0.10 0.34 0.44 0.26 0.28
SAM2-UNet 0.38 0.64 0.83 0.65 0.63 | 0.18 0.45 0.54 0.35 0.38
MedSAM · full fine-tuning 0.21 0.52 0.73 0.54 0.50 | 0.09 0.35 0.45 0.27 0.29
RETFound + UPerNet 0.28 0.59 0.79 0.59 0.56 | 0.12 0.39 0.48 0.30 0.32
AUPR
0.9 0.6 0.3 0.0
Zero-shot and oracle rows use no training image. PEFT and encoder + decoder rows are trained on the source of the test set (internal test). Oracle prompts come from ground-truth components: an upper bound, not a usable setting.
† per-class cells assign each automatic mask the class it overlaps most (upper bound); the bar is the any-lesion AUPR of the class-agnostic map. Density, area cap and SAM 3 wording were chosen on the validation pool.
b
Oracle prompts
SAM SAM 2 MedSAM box point
mAUPR
MA · microaneurysm
SE · soft exudate
AUPR on IDRiD test
nnU-Net, trained
0.9 0.6 0.3 0.0
1 5 20 all
simulated
Oracle prompts per image and class
c
Text prompt wording (SAM 3)
“microaneurysm”
“red dot”
“small red lesion”
“retinal hemorrhage”
“blot hemorrhage”
“dark red patch”
“hard exudate”
“yellow deposit”
“bright lesion”
“cotton wool spot”
“soft exudate”
“white fluffy patch”
simulated
0.0 0.1 0.2 0.3 0.4
AUPR on IDRiD test, zero-shot
d
Does ophthalmic pretraining help?
internal (DDR test) independent, fine (TJDR)
RETFound-DINOv2 ophthalmic, self-sup.
RETFound-MAE ophthalmic, self-sup.
DINOv2 ViT-L natural images, self-sup.
ImageNet-sup. ViT-L natural images, supervised
0.1 0.2 0.3 0.4
mAUPR, same UPerNet head, trained on DDR
e
Best of each kind of use against the references
simulated
DDR test (internal for trained models)
TJDR-std test (independent)
simulated
mAUPR
0.6 0.4 0.2 0.0
0.15 0.11
0.53 0.56
0.38 0.41
0.32 0.28
0.37 0.36
0.43 0.52
Zero-shot SAM 3
Oracle prompt SAM 2
PEFT SAM2-UNet
Encoder + decoder RETFound
Reference M2MRF
Reference nnU-Net